

Transcranial direct current stimulation of primary motor cortex reduces thermal pain

Amin Dehghani^a, David M. Gantz^a, Ethan K. Murphy^b, Michael A. Nitsche^{c,d}, Ryan J. Halter^b, Tor D. Wager^{a,*}

Abstract

Transcranial direct current stimulation (tDCS) is a noninvasive brain stimulation technique with growing potential for pain modulation. However, previous studies have often been limited by small sample sizes, limited outcome reliability, and blinding issues. Pre-registered randomized trials are needed to establish efficacy and reproducibility. Here, we report results from a pre-registered, triple-blind, randomized within-subject crossover trial (N = 50), investigating the effects of left motor cortex (M1) tDCS on experimentally evoked thermal pain in healthy adults. Participants underwent individualized pain calibration, then received anodal, cathodal, and sham M1-tDCS in counterbalanced order on separate visits 4 or more days apart. We assessed post-tDCS thermal pain, potential blinding, side effects, and reaction time during pain rating. Anodal tDCS significantly reduced pain compared with cathodal and sham stimulation ($P < 0.001$, Cohen's $d = 0.93$ and 0.85 , respectively; $P < 0.001$). Cathodal tDCS increased pain relative to sham (Cohen's $d = 0.34$, $P < 0.01$). These effects remained stable throughout the 35-minute post-tDCS pain testing period, with no evidence of decay over time. Blinding assessments confirmed that participants were unable to reliably identify the sham session. Anodal tDCS also modestly increased reaction time in the left (ipsilateral) hand ($P < 0.05$, Cohen's $d = 0.16$), consistent with interhemispheric inhibition, which was uncorrelated with analgesia ($P > 0.10$). This study provides strong evidence for reliable and polarity-dependent modulation of experimental pain by M1-tDCS.

Keywords: tDCS, Pain modulation, Randomized within-subject crossover trial, Left motor cortex (M1), Anodal, Cathodal, Sham, Blinding assessment, Reaction time, Interhemispheric inhibition

1. Introduction

Transcranial direct current stimulation (tDCS) is a noninvasive brain stimulation technique that shows promise as a safe, effective, and scalable treatment option for chronic pain conditions.

Transcranial direct current stimulation involves the delivery of a weak electrical current directly to the surface of the scalp through an electrode array (or montage) to induce subthreshold modulation of neural resting-state membrane potentials.^{55,83} Depending on the montage configuration and polarity, tDCS delivers either excitatory (anodal) or inhibitory (cathodal) stimulation to an area of the cortex underneath the electrode.^{20,69} Transcranial direct current stimulation has been successfully

used in both experimental and clinical studies, including effects on fibromyalgia,^{15,23} migraine,^{12,17,56} neuropathic pain,^{19,63} pelvic pain,^{28,61} and other pain syndromes.^{36,74} In addition to its clinical use, tDCS has also been shown to modulate experimental pain perception in healthy subjects across multiple modalities, including heat, cold, and pressure pain.^{2,21,62,73,75} The most common target in the context of pain modulation using tDCS is the primary motor cortex (M1) and has been found to induce the endogenous release of opioids^{24–26} and modulate the sensory-discriminative aspects of pain.⁵⁸ Repeated daily sessions of tDCS are often used to achieve sustained clinical benefits,^{13,32,80} but previous studies have shown that single-session tDCS can induce short-term effects on pain perception in healthy individuals.^{22,31,37,39,41,45,49,53,58,62,68,82,88}

Although M1-tDCS has been shown to modulate experimental pain perception, there remains substantial uncertainty about how robust its effects are and whether they can be partly attributable to confounds such as ineffective blinding or expectation-related biases.²¹ Findings across studies have been variable, and effect sizes are generally modest (overall Hedges' g of 0.2^{58}), with several likely contributing factors. Many previous studies relied on static pain protocols to measure pain and tolerance across a small number of stimulation trials^{10,11,31,38,39}; included small sample sizes (generally < 30),^{22,31,37,39,41,45,49,53,62,68,82,88} and did not rigorously assess or account for blinding.^{9,38,57,60} Moreover, the use of tDCS can induce strong expectation effects,^{40,78} which may mask the true effects of stimulation if not adequately addressed. Therefore, larger, more rigorous studies are necessary to confirm polarity-specific effects and address limitations related to reproducibility and methodological consistency.

Sponsorships or competing interests that may be relevant to content are disclosed at the end of this article.

^a Department of Psychological and Brain Sciences, Dartmouth College, Hanover, NH, United States, ^b Thayer School of Engineering and Geisel School of Medicine, Dartmouth College, Hanover, NH, United States, ^c University Clinic of Psychiatry and Psychotherapy, Protestant Hospital of Bethel Foundation, University Hospital OWL, Bielefeld University, Bielefeld, Germany, ^d German Center for Mental Health (DZPG), Bochum, Germany

*Corresponding author. Address: Department of Psychological and Brain Sciences, HB 6207, Dartmouth College, Hanover, NH 03755, United States. Tel.: 603-646-2196. E-mail address: tor.d.wager@dartmouth.edu (T. D. Wager).

Copyright © 2025 The Author(s). Published by Wolters Kluwer Health, Inc. on behalf of the International Association for the Study of Pain. This is an open access article distributed under the terms of the Creative Commons Attribution-Non Commercial-No Derivatives License 4.0 (CCBY-NC-ND), where it is permissible to download and share the work provided it is properly cited. The work cannot be changed in any way or used commercially without permission from the journal.

<http://dx.doi.org/10.1097/j.pain.0000000000003851>

This pre-registered,³³ triple-blind, within-subject crossover randomized controlled trial (RCT) evaluated the effects of left M1-tDCS (anodal, cathodal, sham) on thermal pain responses on the contralateral right forearm in $N = 50$ healthy participants. Pain stimuli were individually calibrated to each participant's pain and tolerance thresholds, and participants completed pain testing before and after each stimulation session. Based on prior work,²¹ we hypothesized that anodal M1-tDCS would reduce pain ratings relative to cathodal and sham, and that cathodal stimulation would increase pain ratings relative to sham.

2. Materials and methods

2.1. Participants

Seventy healthy participants were recruited from Dartmouth College and the surrounding New Hampshire community. Exclusion criteria were younger than 18 years or older than 55 years, chronic pain, current or history of major psychiatric or neurological diagnosis, substance abuse in the past 6 months, or having contraindications for brain stimulation or experimental pain (eg, pregnancy). To preserve blinding—particularly given the noticeable differences between sham and active tDCS—individuals affiliated with the Department of Psychological and Brain Sciences at Dartmouth College (eg, students, staff, and faculty) were excluded from participation. All participants were also naïve to tDCS and brain stimulation, helping to reduce expectancy effects and enhance blinding integrity. The study was approved by the Dartmouth College Institutional Review Board, and informed consent was obtained from all participants before every session, in accord with the Declaration of Helsinki.

Based on the findings from a previous study,²¹ which reported a large effect size (Cohen's $d = 0.86$) for the contrast between anodal and cathodal tDCS in pain modulation, we estimated a medium effect size ($d = 0.43$) or larger for the contrast between anodal and sham stimulation in the current design. Assuming a one-tailed test at $\alpha = 0.025$, a sample size of $N = 50$ was required to detect this effect with 85% power. We initially recruited 70 participants. Fourteen participants were excluded after pain calibration (see below), and 6 participants dropped out of the study during or after the first tDCS session (one participant discontinued due to discomfort from the painful stimuli, and 5 participants dropped out after completing the first tDCS session), leaving a final sample size of $N = 50$ (25 biological women, $M_{\text{age}} = 26.9 \pm 9.2$, range: 18–55 years). The study design is presented in **Figure 1**.

This participant sample also completed additional tasks (completed after the main task presented here) conducted as part of a broader experimental protocol. No data from this study have been previously published, but we are planning to publish data from these additional tasks in subsequent articles.

2.2. Adaptive thermal calibration procedure (session 1)

Eligible participants were invited to participate in an initial adaptive thermal calibration procedure. This method improves measurement accuracy by adjusting temperature based on participants' sensation reports, allowing individual pain and tolerance levels to be determined. The procedure involved participants measuring 3 finger widths proximal from the wrist crease and 4 finger widths distal from the elbow crease. The midpoint between these 2 landmarks was identified, and 3 sites were marked along the central longitudinal axis of the forearm: one at the midpoint, one at 1 cm above, and one at 1 cm below. Two additional sites were marked by drawing diagonal lines from the upper wrist point to the lower elbow point and from the lower wrist point to the upper

elbow point, with the intersections forming 2 lateral stimulation sites, resulting in 6 total locations (**Fig. 1**).

The calibration consisted of 4 runs (24 trials total). The first run (6 trials) served as a familiarization phase and data were not included in the data analysis. Each noxious heat stimulus lasted 13 seconds in total, including a 1.5-second ramp-up, a 10-second plateau at target temperature, and a 1.5-second ramp-down. Noxious heat stimuli were applied to 6 forearm sites. After each individual stimulation, participants answered 3 questions. First, they rated sensation intensity using a semicircular generalized labeled magnitude scale (gLMS),⁵ ranging from “No sensation” (0°) to “Barely detectable” (3° of scale angle), “Weak” (10°), “Moderate” (29°), “Strong” (64°), “Very Strong” (98°), and “Strongest sensation of any kind” (180°). Then, they provided binary yes/no responses to 2 questions: (1) “Was the sensation painful?” and (2) “Is this level of stimulation tolerable for the duration of the experiment?”. The initial run familiarized participants with 47°C heat stimuli applied to each of 6 skin sites. In run 2, 6 stimuli were presented at fixed temperatures (2 each at 45°C, 47°C, and 48.9°C) to train initial regression models to estimate pain levels, pain threshold, and pain tolerance. Two logistic regression models were used to identify pain threshold and tolerance. The first model used “painful yes/no” binary responses as the outcome and predicted the likelihood of pain as a function of temperature. The second logistic regression model used “tolerable yes/no” binary responses as the outcome as a function of temperature. In runs 3 and 4 (6 trials each), we re-ran the logistic regression models after each trial, and adjusted the temperature on the next trial adaptively based on the predicted responses. Stimuli were randomly assigned to “low” and “high” categories (3 trials each per run). For a “low” stimulus, the temperature chosen was the temperature with a fitted probability of being above the pain threshold of 0.9 (ie, $P_{\text{pain}} = 0.9$) from the logistic regression applied to all previous trials. For a “high” stimulus, the next temperature was the temperature with a fitted probability of being below the tolerance threshold of 0.9 (ie, $P_{\text{tolerable}} = 0.9$) from the logistic regression applied to all previous trials. In this way, the adaptive algorithm results in sampling most densely from temperatures just above or below pain and tolerance thresholds, providing a successively better approximation of these thresholds on each calibration trial. After all calibration trials, a linear regression model was used to identify the predicted pain intensity as a function of temperature. The 3 skin sites with the lowest mean absolute error in predicted pain intensity were selected for stimulation in subsequent sessions. This procedure helps ensure that all skin sites tested are approximately equally pain sensitive. Subsequent experimental tests balanced stimulation on all 3 sites equally in every experimental condition. For additional details on this adaptive pain calibration, see Ref. 21.

Participants with less than a 1°C difference between pain and tolerance were excluded. Seventy participants completed the calibration session, and 56 participants passed and were selected for inclusion in the tDCS sessions.

2.3. Experimental sessions (sessions 2–4)

Participants returned for 3 additional sessions on separate days, with a minimum 4-day interval to minimize carryover effects of tDCS (13.6 ± 11.7 , range: 4–75 days between sessions), consistent with previous studies using multiday washout periods in crossover designs.^{1,6,7} In each session, participants underwent 3 eight-minute pain tests at 3 different skin sites on the right forearm (pre-tDCS, 12 trials per skin site, 36 trials total), followed by 20 minutes of continuous tDCS, a questionnaire about the tDCS experience, and finally 3 eight-minute pain runs at

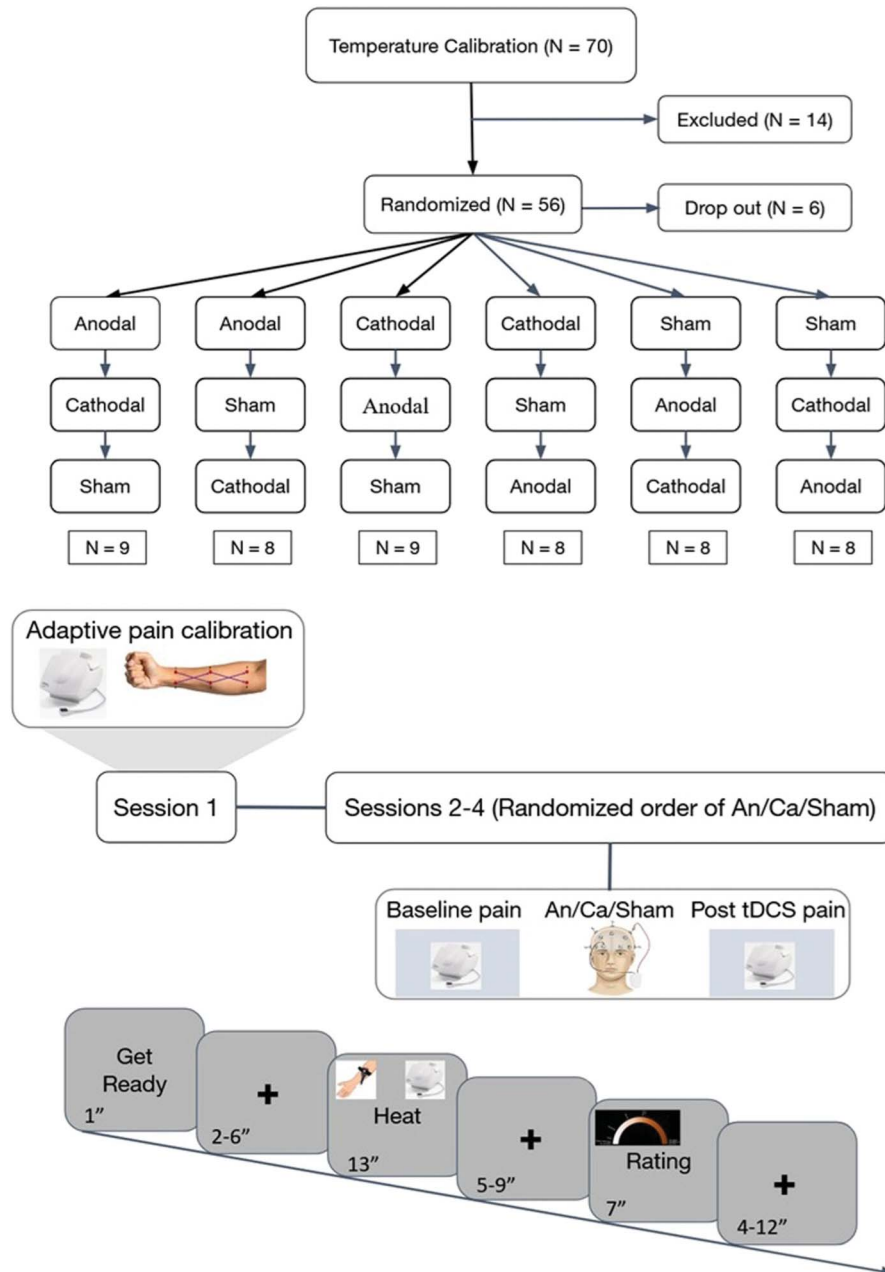


Figure 1. Overview of paradigm, design, and pain tests.

3 different skin sites on the right forearm (post-tDCS, 12 trials per skin site, 36 trials total). The 3 runs pre- and post-tDCS were conducted back-to-back, with no planned interval between them; the only pause was the ~30 seconds required to move and secure the thermode to a new skin site. TDCS was either left anodal M1, cathodal M1, or sham M1 stimulation, with participants randomly assigned to a treatment order (Fig. 1).

2.4. Noxious heat stimulation

Noxious heat was delivered randomly to 3 locations on the right forearm using a TSA-II Neurosensory Analyzer (Medoc Ltd) with a 16-mm thermode end-plate. Participants received three 8-minute runs of repeated stimulation before and after tDCS stimulation (6 runs total). Each run included 12 trials with 6

randomly interleaved, individually calibrated low and 6 high temperatures (see Results for intensities) delivered to the right volar forearm. Each trial lasted for 40 seconds and started with a warning cue (the words “Get Ready”; 1 second), followed by a 2 to 6 seconds delay, heat stimulation (13 seconds including 1.5 seconds ramp-up/down, 10 seconds plateau), a 5 to 9 seconds delay, a pain rating period (7 seconds using the left hand), and a final 4 to 12 seconds delay before the next cue. Participants rated the sensation using the semicircular gLMS with their left hand.

2.5. Transcranial direct current stimulation design

Transcranial direct current stimulation was delivered using a StarStim 8 system (Neuroelectronics, Spain), which delivered 2

mA of stimulation through saline-soaked circular sponge electrodes (25 cm² each) for 20 minutes. A 30-second ramp-up and ramp-down period was included to minimize stimulation-related discomfort. The impedance of stimulation channels was monitored before and during each session and remained below 10 k Ω for all participants, ensuring safe and uninterrupted stimulation delivery. Electrodes were placed over the C3 (the anode for anodal M1 tDCS) and FP2 (the cathode for anodal M1 tDCS) regions according to the 10-10 system. In the sham stimulation condition, the current was ramped up over 30 seconds, maintained for 1 minute, and then gradually ramped down to mimic the sensations of active stimulation. To preserve blinding, participants were randomly assigned to receive either anodal sham (C3 as anode, FP2 as cathode) or cathodal sham (FP2 as anode, C3 as cathode), with half allocated to each configuration.

After tDCS, participants completed a brief questionnaire assessing sensations commonly induced by tDCS. Using the same gLMS described earlier, participants rated the intensity of itching, tingling, metallic-iron taste, pain, pleasantness or unpleasantness, and warmth-heat.²¹ These ratings were used to evaluate the subjective experience of tDCS-induced sensations, and to determine whether sensory side effects differed across stimulation conditions (anodal, cathodal, or sham). This helped rule out differences in perceived sensation as potential confounds contributing to the observed effects on pain modulation. In addition, to reduce dimensionality and identify underlying sensory dimensions, we conducted a factor analysis on the 6 sensation ratings. The resulting factors were subsequently entered into sensitivity models to examine whether individual differences in perceived stimulation sensations influenced poststimulation pain ratings.

After the last session, participants underwent a funneled debriefing. They were first asked to guess whether any of the sessions was active or sham. Then, they were informed that one of the sessions was sham and asked to guess which it was. These 2 questions were asked to assess participants' blinding to the type of tDCS received, as well as to evaluate the sensations induced by tDCS.

This study used a triple-blind design in which participants, experimenters, and data analysts were all unaware of the stimulation condition. Each participant was assigned to one of 6 pr randomized stimulation sequences shown in **Figure 1**, generated in advance by a researcher not involved in data collection or analysis. During each session, the experimenter entered only the participant number and session number into a custom MATLAB program, which automatically applied the corresponding stimulation (anodal, cathodal, or sham) without revealing the underlying condition. The assigned stimulation was coded with anonymous session labels (eg, sessions A, B, and C), and the data provided to the analyst were fully deidentified and labeled only with participant IDs and blinded session codes. Thus, blinding of the analyst was achieved, ensuring that participants, experimenters, and data analysts all remained blinded to the stimulation condition throughout the study.

2.6. Analysis model

The study included a 3 $\times$ 2 experimental design (tDCS type $\times$ Stimulus Intensity: anodal, cathodal, and sham $\times$ low and high temperature from calibrated pain and tolerance thresholds). We tested the main effects of each factor and their interaction, with planned contrasts capturing differences across tDCS conditions (Anodal > Cathodal & Sham, Cathodal > Sham), as well as the durability of tDCS effects over time, potential effects of testing

order, tDCS sensations, and other effects described below. Sensation ratings that deviated more than 3 standard deviations from a participant's mean value within each of the 6 cells in the 3 $\times$ 2 design were excluded. Less than 1% of trials were excluded based on this criterion.

The statistical model was a mixed-effects model with random participant intercepts and random slopes for tDCS type, Stimulus Intensity, and tDCS type $\times$ Stimulus Intensity (low and high temperature). This model was implemented using the "fitlme" function in MATLAB R2024b (MathWorks, Natick, MA) with maximum likelihood parameter estimation. For this purpose, full trial-level data were included as well as covariates including pretreatment mean pain scores and treatment order groups (6 groups; **Fig. 1**). Biological sex and age were included as between-person covariates. We considered additional covariates for sensations elicited by tDCS type (eg, itching, tingling, metallic-iron taste, pain, warmth, and pleasantness), but anodal, cathodal, and sham tDCS produced equivalent self-reported sensations (all $P > 0.25$), so we did not include these covariates in the main model. The Satterthwaite method was used to estimate the degrees of freedom. In addition, to screen for baseline biological sex differences, we compared calibrated pain and tolerance thresholds between biological women and biological men using independent-samples Welch t-tests. The mixed-effects model used in this study is summarized as:

$$\begin{aligned} \text{Pain} - \text{Post} \sim & 1 + \text{Biological Sex} + \text{Age} + \text{Noxious Heat} \\ & + \text{Pain-Pre} + \text{Days} \times \text{Runs} \times \text{Trials} + \text{Stim Order} \\ & + (\text{tDCS} - \text{An vs. Other} + \text{tDCS} - \text{Ca vs. Sham}) \\ & \times (\text{Runs} \times \text{Trials} + \text{Stimulus Intensity}) \\ & + (\text{Days} \times \text{Runs} \times \text{Trials}) \\ & + (\text{tDCS} - \text{An vs. Other} + \text{tDCS} - \text{Ca vs. Sham}) \\ & \times (\text{Runs} \times \text{Trials} + \text{Stimulus Intensity}) | (\text{Participant}) \end{aligned} \quad (1)$$

In equation 1, Pain-Post is the pain rating on each trial after tDCS treatment. Transcranial direct current stimulation-An vs Other represents the effect of anodal vs average effect of cathodal and sham stimulation (coded with [2/3, -1/3, -1/3] for anodal, cathodal, and sham), tDCS-Ca vs Sham represents cathodal vs sham stimulation (coded with [0, 1/2, -1/2] for anodal, cathodal, and sham). Pain-Pre is the mean pain rating before tDCS in each session. Stim Order refers to 6 different stimulation orders across 3 tDCS sessions (coded with integers 1-6 for the different order types, see **Fig. 1** for 6 different orders for 3 tDCS sessions). Days, Runs, and Trials refer to linear effects of session (testing day, 3 in total, mean-centered), run number (run within session across 3 runs, mean-centered within day), and trial number (trial within run across 12 trials, mean-centered within run), respectively. These variables were included to control for habituation and/or sensitization across time. Age and Biological Sex refer to the age and biological sex of each participant, Stimulus Intensity represents the temperature obtained by calibration (low and high temperature based on pain and tolerance threshold, coded with 1 and -1 for high and low), Noxious Heat refers to the temperature of the applied stimulus, and Participant represents the random effect of the participant. In this model, the interactions between tDCS type and Runs and tDCS type and Trials within run test the effect durability of tDCS over the 35-min poststimulation test period. Anodal vs sham effects were tested using a separate mixed model focusing on this comparison, including other factors and covariates as above. In addition, a simplified model without interaction terms was also tested to assess the robustness of the tDCS effects.

To compare the reaction time before and after tDCS, a model similar to equation (1) was used, as follows:

$$\begin{aligned} \log(\text{RT} - \text{Post}) \sim & 1 + \text{Biological Sex} + \text{Age} + \text{Noxious Heat} \\ & + \log(\text{RT} - \text{Pre}) + \text{Days} \times \text{Runs} \times \text{Trials} + \text{Stim Order} \\ & + (\text{tDCS} - \text{An vs. Other} + \text{tDCS} - \text{Ca vs. Sham}) \\ & \times (\text{Runs} \times \text{Trials} + \text{Stimulus Intensity}) + (\text{Days} \times \text{Runs} \\ & \times \text{Trials} + [\text{tDCS} - \text{An vs. Other} + \text{tDCS} - \text{Ca vs. Sham}] \\ & \times [\text{Runs} \times \text{Trials} + \text{Stimulus Intensity}] | \text{Participant} \end{aligned} \quad (2)$$

In equation (2), RT-Post is the reaction time of pain ratings after tDCS and RT-Pre is the reaction time of pain ratings before tDCS. The remaining variables used in equation (2) are similar to those ones used in equation (1).

A few minor deviations from the pre-registered analysis plan were made. Age was added as a covariate to improve model precision. Sensory ratings (eg, itching, tingling, warmth) were not included in the primary model because they did not significantly differ across stimulation conditions, but were examined in a follow-up sensitivity model. Finally, we analyzed reaction time to assess potential motor effects of tDCS, although this outcome was not included in the pre-registration.

3. Results

3.1. Calibration

The mean $\pm$ standard deviation values for pain and tolerance thresholds were $46.4 \pm 0.7^\circ\text{C}$ and $48.2 \pm 0.7^\circ\text{C}$, respectively, and a within-person difference $1.8 \pm 0.8^\circ\text{C}$ (Fig. 2 for the distribution of low and high temperatures). This study recruited an equal number of participants from both biological sexes, and there was no significant difference in the mean values for pain and tolerance between biological men and biological women ($P > 0.05$).

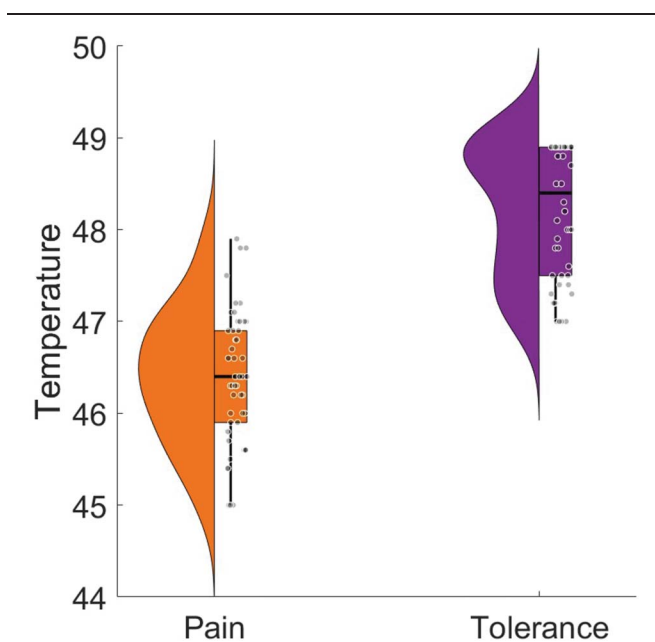


Figure 2. Calibration results. Boxplots indicate the median, 25th, and 75th percentiles, and individual data points.

3.2. Sensations caused by transcranial direct current stimulation

In sessions 2 to 4, after each tDCS session, participants rated the sensations evoked by stimulation. Ratings included itching, tingling, metallic/iron taste, pleasantness or unpleasantness, warmth/heat, and pain (Fig. 3). A one-way repeated-measures ANOVA comparing anodal, cathodal, and sham stimulation sessions revealed no statistically significant differences in reported sensations across stimulation types (all P -values > 0.25), indicating no evidence that the type of stimulation systematically influenced sensory experiences.

To further examine the structure of these subjective sensations, factor analysis on the 6 rated dimensions resulted in 2 latent factors. The first factor, characterized by strong positive loadings for pain and warmth-or-heat and a negative loading for pleasantness, likely represents a scalp pain-and-heat sensory dimension, where higher pain and thermal sensations are associated with lower pleasantness. The second factor, with high positive loadings for itching and tingling, seems to reflect an itching-tingling sensory dimension, capturing cutaneous and paresthetic sensations. These factors were used in a secondary model to evaluate whether sensory experience moderated the effects of tDCS on pain ratings.

3.3. Transcranial direct current stimulation effects on pain

As shown in Figure 4 and Table 1, Pain was significantly reduced for anodal relative to the average effect of cathodal and sham stimulations (a main effect of tDCS-An vs Other, $F(1,36.4) = 52.1$, $P < 0.001$, Cohen's $d = 0.93$ for anodal vs cathodal and Cohen's $d = 0.85$ for anodal vs sham) and significantly increased for cathodal relative to sham (a main effect of tDCS-Ca vs Sham, $F(1,45) = 9.9$, $P < 0.01$, Cohen's $d = 0.34$). As expected, pain was higher with higher Stimulus Intensity (a main effect of Stimulus Intensity, $F(1,50.0) = 183.2$, $P < 0.001$). We found no significant effect for Noxious Heat ($F(1,34.2) = 2.5$, $P = 0.12$). There were no tDCS type $\times$ Stimulus Intensity interactions (see Table 1 for statistics).

As expected, post-treatment pain was higher when pre-treatment pain was higher on the same day, reflecting day-to-day variability (Pain-Pre, $F(1,82.4) = 410.4$, $P < 0.001$). Pain increased significantly across Days ($F(1,48.1) = 4.2$, $P < 0.05$) and Runs within day ($F(1,49.8) = 4.2$, $P < 0.05$), reflecting sensitization effects observed in some previous studies,⁴⁸ but not significantly across Trials within-run ($F(1,49.1) < 0.001$, $P = 0.99$). We found no main effects of treatment order (Stim Order; $F(5,30.9) = 2.1$, $P = 0.08$), consistent with a lack of carry-over across sessions.

Between-person demographic covariates indicated that pain was lower for biological men relative to biological women ($F(1,34.7) = 4.7$, $P < 0.05$), a frequent but not ubiquitous finding in prior literature,³⁰ and lower in older participants ($F(1,34.4) = 12.8$, $P < 0.01$).

Transcranial direct current stimulation effects were stable across the testing session and did not diminish across the 35-min post-tDCS testing period, as suggested by nonsignificant interactions between tDCS-An vs Other or tDCS-Ca vs Sham and Runs within day or Trials within run. Thus, the effects of 20-minute tDCS did not “wear off” over time. In addition, no differences were observed between anodal-sham (C3 anode and FP2 cathode) and cathodal-sham (FP2 anode and C3 cathode) configurations in poststimulation pain ratings (all $P > 0.30$), indicating that the 2 sham montages were equivalent. Full fixed effect coefficients, F statistics, error degrees of freedom, and

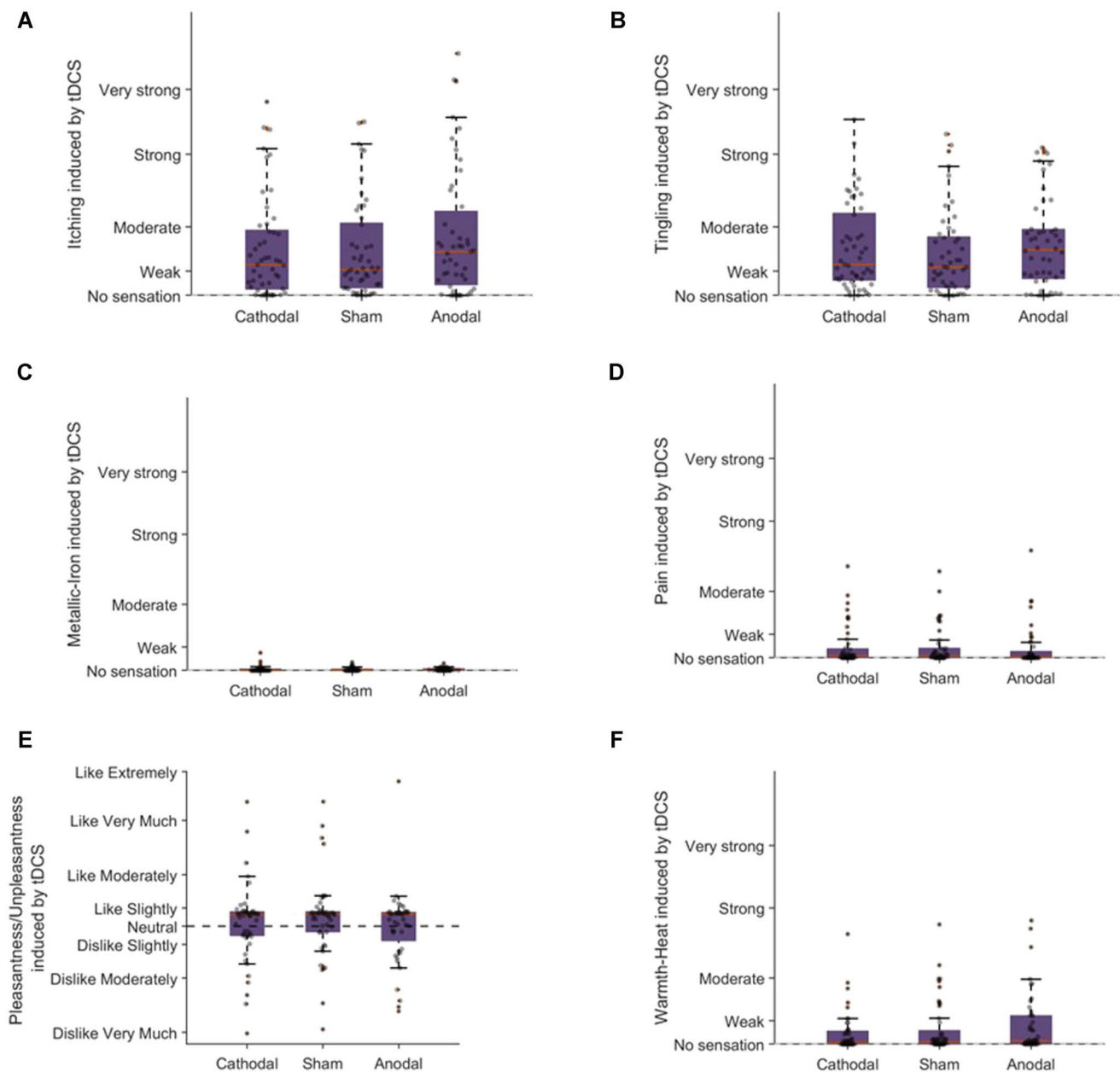


Figure 3. Sensations caused by anodal, cathodal, and sham tDCS. (A–F): no significant difference between the sensations (itching, tingling, metallic-iron, pain, pleasantness/unpleasantness, and warmth-heat) evoked by each type of tDCS ($P > 0.09$). Boxplots indicate the median, 25th, and 75th percentiles, and individual data points.

P -values for the main model (equation 1) are summarized in **Table 1**. Violin plots for effects of tDCS Type (anodal vs cathodal, anodal vs sham, and cathodal vs sham comparisons), pain rating change (Post - Pre) for high and low temperature, and pain ratings before and after tDCS for high and low temperature are shown in **Figure 4**.

To visualize how pain responses evolved over time during the test session across tDCS conditions, we plotted model-corrected pre- and poststimulation pain ratings across the 3 runs and 12 trials within each run for cathodal, sham, and anodal stimulation (**Fig. 4F–G**). For poststimulation ratings, the model adjusted for baseline pain ratings (Pain-Pre), Days, Stimulus Intensity, Noxious Heat, Stim Order, Biological Sex, and Age as fixed effects, with subject-level random intercepts and slopes. Prestimulation ratings were corrected using the same model structure, excluding Pain-

Pre as a covariate. We observed a sharp drop in pain ratings from the first to the second trial across all 3 runs and in both pre- and post-tDCS sessions. This immediate decrease likely reflects a combination of rapid sensory adaptation and central habituation processes, consistent with previous findings.^{42,44,65}

3.4. Sensitivity analyses to examine robustness of transcranial direct current stimulation effects

We conducted several types of sensitivity analysis to test whether the main findings remained significant across different modeling choices. A simplified mixed-effect model with only the main effects and interaction, but without additional covariates, showed the same pattern of effects: The critical tDCS-An vs Other and tDCS-Ca vs Sham effects remained significant (both $P < 0.001$).

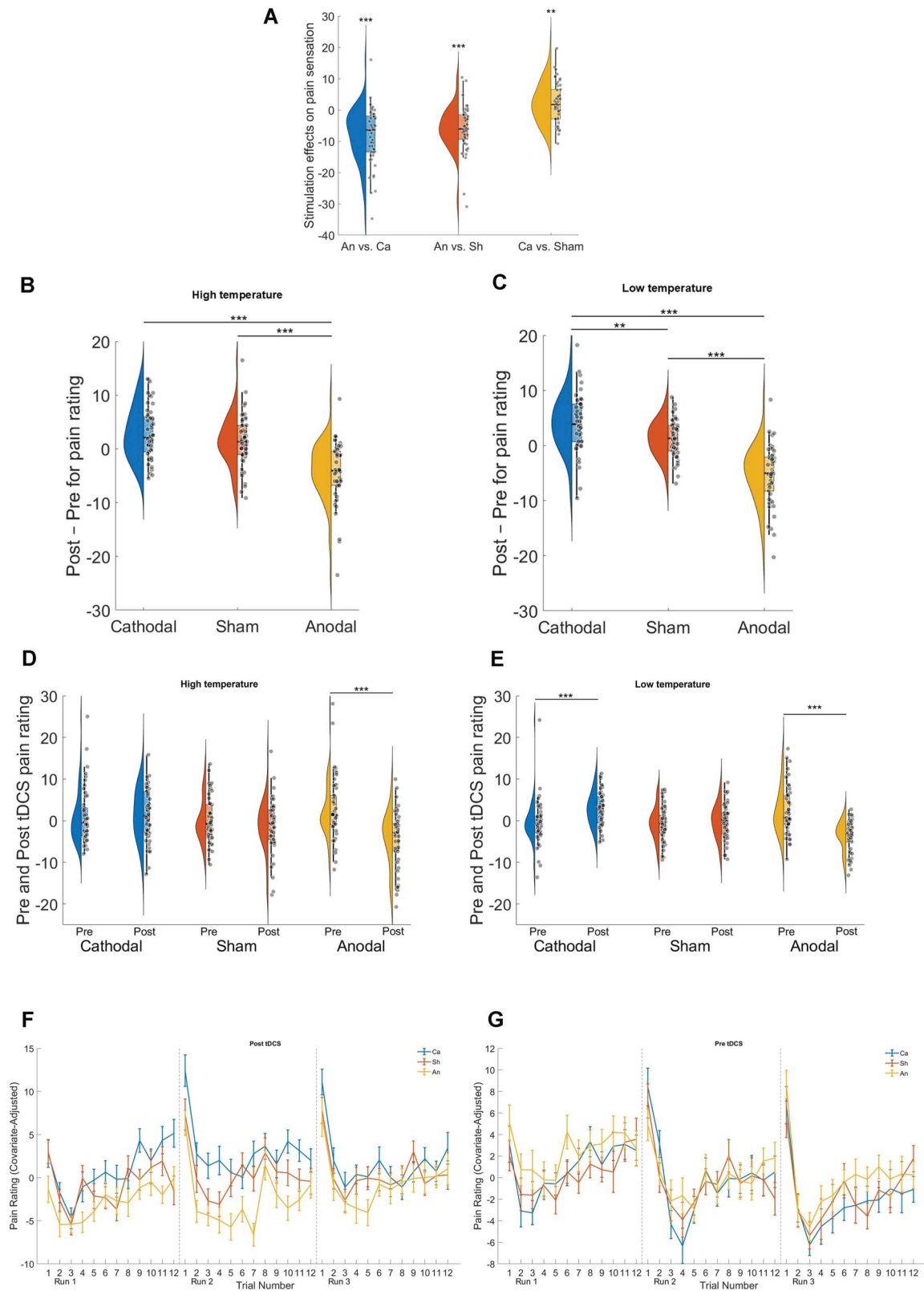


Figure 4. Main findings for pain sensation. (A) The effects of anodal vs cathodal, anodal vs sham, and cathodal vs sham stimulation on pain sensation. (B–C) Treatment effects (Post-treatment—Pretreatment pain rating change) for each of the 3 stimulation types and for each of high temperature (left panel) and low temperature (right panel) heat. (D–E) Pain ratings pre and post-treatment for each of the 3 stimulation types and for each of high-temperature (left panel) and low-temperature (right panel) heat. (F) Model-corrected prestimulation pain ratings across 3 runs and 12 trials within each run for anodal, cathodal, and sham stimulations. Corrections removed variance from nuisance covariates (eg, Days, Stimulus Intensity, Noxious Heat, Stim Order, Age, and Biological Sex) using a simplified mixed-effects model. (G) Model-corrected poststimulation pain ratings across 3 runs and 12 trials within each run for anodal, cathodal, and sham stimulations. Corrections removed variance from nuisance covariates (eg, Pain-Pre, Days, Stimulus Intensity, Noxious Heat, Stim Order, Age, and Biological Sex) using a simplified mixed-effects model. This isolates stimulation-related effects on pain trajectories. Boxplots indicate the median, 25th, and 75th percentiles, and individual data points. Dots reflect effect magnitude estimates of individual participants. All values were adjusted to remove participant-level intercepts. *** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$ two-tailed.

Table 1**Mixed-effects model results for pain perception after tDCS.**

Variable	Estimate	F-stat	DF1	DF2	P
Intercept	−50.48	1.9	1	34.1	0.17
Days*	0.72	4.2	1	48.1	<0.05
Runs*	0.71	4.2	1	49.8	<0.05
Stimulus intensity†	9.77	183.2	1	50.0	<0.001
Noxious heat	1.23	2.5	1	34.2	0.12
Trials	5.81e-04	<0.001	1	49.1	0.99
Pain-Pre‡	0.72	410.4	1	82.4	<0.001
tDCS-An vs other†	−6.39	52.1	1	36.4	<0.001
tDCS-Ca vs Sham‡	2.48	9.9	1	45.0	<0.01
Stim order§	n.a	2.1	5	30.9	0.08
Biological sex—M vs F*	−2.00	4.7	1	34.7	<0.05
Age‡	−0.19	12.8	1	34.4	<0.01
Days:runs	0.29	0.5	1	31.3	0.49
Days:trials	2.20e-02	0.2	1	115.6	0.66
Runs:trials†	−0.25	13.8	1	54.4	<0.001
Runs: tDCS-An vs other	0.24	0.1	1	39.5	0.72
Stimulus intensity: tDCS-An vs other	0.33	0.4	1	50.8	0.52
Trials: tDCS-An vs other	9.57e-02	1.2	1	136.0	0.27
Runs: tDCS-Ca vs Sham	−0.12	<0.1	1	19.4	0.86
Stimulus intensity: tDCS-Ca vs Sham	−0.24	0.1	1	50.2	0.70
Trials: tDCS-Ca vs Sham	1.73e-02	<0.1	1	91.2	0.86
Days:runs:trials	0.11	2.3	1	74.3	0.13
Runs:trials:tDCS-An vs other	7.80e-02	0.3	1	53.3	0.58
Runs:trials:tDCS-Ca vs Sham	−0.17	1.5	1	54.1	0.22

* $P < 0.05$ two-tailed.† $P < 0.001$ two-tailed.‡ $P < 0.01$ two-tailed.

§ n.a., not applicable.

For Stim Order, a categorical variable with 6 levels, only the overall F-test and P -value are reported; no single estimate is shown.

":", means interaction between variables.

A second model tested potential interactions with tDCS sensations, which could potentially alter expectations and thus affect pain ratings, as has been found with previous studies of side effects.⁷⁹ The 2 sensory dimensions identified through factor analysis were included as predictors, along with their interactions with tDCS type in the mixed-effects model in equation (1). The effects of tDCS-An vs Other and tDCS-Ca vs Sham remained significant (both $P < 0.001$), indicating that the observed effects on pain were not attributable to individual differences in sensory experience across stimulation conditions.

A third model tested the potential interaction between tDCS and Biological Sex and Age, and the results showed the effect of tDCS-An vs Others and tDCS-Ca vs Sham effect remained significant ($P < 0.001$ and $P < 0.01$, respectively), without any significant interaction with Biological Sex and Age ($P > 0.64$ and $P > 0.15$, respectively).

A fourth model tested the effect of anodal vs Sham (tDCS-An vs Sham) and cathodal vs average of anodal and sham (tDCS-Ca vs Other). The results showed that pain was significantly reduced for anodal relative to sham stimulation (a main effect of tDCS-An vs Sham, $F(1,35.1) = 41.1$, $P < 0.001$) and significantly increased for cathodal relative to the average effect of anodal and sham

stimulations (a main effect of tDCS-Ca vs Other, $F(1,42.7) = 33.3$, $P < 0.001$).

Finally, to assess potential randomization failures, a one-way repeated-measures ANOVA was conducted on the average pre-tDCS pain ratings across the 3 stimulation sessions. The analysis revealed no significant differences in baseline pain ratings among sessions for either the low-temperature stimuli ($F(2,147) = 1.2$, $P = 0.31$) or the high-temperature stimuli ($F(2,147) = 0.4$, $P = 0.64$). These results suggest that randomization was adequate and there were no group differences at baseline.

We have examined effects on pain across 3 time scales and observed: (1) effects of day, with a modest increase in pain across days; (2) effects of run within day, with a modest increase in pain across runs within a day; and (3) effects of trial within run, with a drop in pain after the first trial followed by a gradual increase. Trial effects across 3 runs are shown in **Figure 4F**. This study was designed so that the effects of day, run, and trial on pain are not confounded with stimulation effects, because stimulation effects are fully orthogonal to these variables by design. The effects of tDCS stimulation remained significant whether we controlled for these variables and did not interact with them. In additional analyses, we also found that tDCS effects remained significant

after eliminating the first trial of each run (where the largest drop occurred). Therefore, changes in pain across time do not affect the generalizability of our stimulation findings.

3.5. Assessments of blinding

Analyses of debriefing questions indicated that participants could not discriminate active from sham tDCS. Thirty-eight of 50 participants (76%) initially guessed that all conditions were active. When informed that exactly one session was sham and asked to identify which of the 3 sessions was the sham, participants performed significantly worse than chance, with 12 of 50 (24%) guessing correctly. Thus, any expectancy effect would thus be expected to favor sham.

3.6. Effects on reaction time

The results of the reaction time analysis are presented in **Figure 5** and **Table 2**. Reaction time (RT) significantly increased for anodal relative to cathodal and sham stimulations (main effect of tDCS-An vs Other, $F(1,47.6) = 4.1$, $P < 0.05$, Cohen's $d = 0.16$ for anodal vs cathodal and sham). This could indicate an effect of tDCS on either motor function or confidence in pain ratings. RT was also lower with higher Stimulus Intensity (see **Table 2** for statistics). As expected, post-treatment RT covaried with pre-treatment RT on the same day (RT-Pre, $F(1,63.3) = 992.8$, $P < 0.001$). RT also decreased significantly across Runs within day ($F(1,55.2) = 5.0$, $P < 0.05$), indicating a practice effect, but not significantly across Trials within-run ($F(1,62.0) = 2.9$, $P = 0.09$).

Full fixed effect coefficients, F statistics, error degrees of freedom, and P values for the main model (equation 2) are summarized in **Table 2**, and violin plots for the tDCS type (anodal vs cathodal, anodal vs sham, and cathodal vs sham comparisons) and reaction time change (Post-Pre) for high and low temperature are shown in **Figure 5**.

There was no significant correlation between reaction time slowing and the analgesic effects of tDCS ($P > 0.10$), suggesting that reaction time changes were independent of pain modulation.

4. Discussion

This study used a pre-registered, triple-blind, randomized, within-subject crossover study including 4 separate sessions to investigate the effects of left M1-tDCS on thermal pain modulation in healthy adults. Session 1 adaptively calibrated thermal intensities, establishing each participant's pain and tolerance thresholds for low- and high-intensity stimuli in the tDCS sessions. This was followed by 3 tDCS sessions—anodal, cathodal, and sham—administered in a counterbalanced order with a minimum 4-day interval between sessions to minimize carryover effects. Unlike many previous studies that used small sample sizes,^{2,22,31,37,39,41,45,49,50,53,62,67,68,82,88} used limited numbers of pain trials, and relied on static pain protocols,^{10,11,31,38,39} this study incorporated repeated pain trials with randomized stimulus intensities, potentially reducing bias and demand characteristics, both before and after tDCS. Demand characteristics refers to features of the experimental context that may lead participants to infer the study's hypotheses or expected

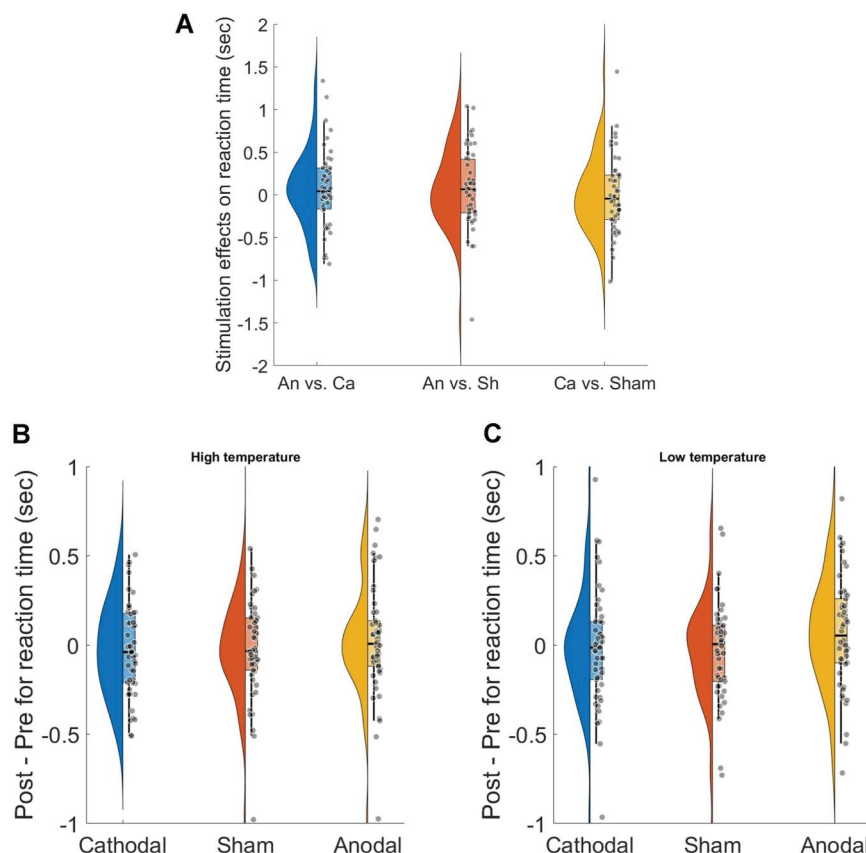


Figure 5. Main findings for reaction time. (A) The effects of anodal vs cathodal, anodal vs sham, and cathodal vs sham stimulation on reaction time. (B–C) Treatment effects (Post-treatment—Pretreatment reaction time change) for each of the 3 tDCS types, and for each of high temperature (left panel) and low temperature (right panel) heat. Boxplots indicate the median, 25th, and 75th percentiles, and individual data points. Dots reflect effect magnitude estimates of individual participants. All values were adjusted to remove participant-level intercepts. *** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$ two-tailed.

Table 2**Mixed-effects model results for reaction time after tDCS.**

Variable	Estimate	F-stat	DF1	DF2	P
Intercept	0.19	0.1	1	53.8	0.74
Days	1.69e-02	2.6	1	50.3	0.11
Runs*	−1.24e-02	5.0	1	55.2	<0.05
Stimulus intensity†	−1.73e-02	14.3	1	54.8	<0.001
Noxious heat	−1.23e-03	<0.1	1	53.5	0.92
Trials	−2.18e-03	2.9	1	62.0	0.09
RT-Pre‡	0.85	992.8	1	63.3	<0.001
tDCS-An vs other*	3.69e-02	4.1	1	47.6	<0.05
tDCS-Ca vs. Sham	1.57-03	<0.01	1	45.3	0.94
Stim order‡	n.a	1.0	5	48.7	0.44
Biological sex—M vs F	2.06e-03	<0.1	1	55.6	0.88
Age	3.62e-04	0.2	1	51.8	0.65
Days:Runs	9.42e-03	2.3	1	74.9	0.13
Days:trials*	−4.05e0-3	6.5	1	53.9	<0.05
Runs:trials	−1.84e-04	<0.1	1	80.9	0.89
Runs: tDCS-An vs other	−1.96e-03	<0.1	1	96.4	0.84
Stimulus intensity: tDCS-An vs other	−5.63e0-3	0.5	1	93.4	0.50
Trials: tDCS-An vs other	−1.72e-04	<0.01	1	69.8	0.94
Runs: tDCS-Ca vs Sham	−1.88e-03	<0.1	1	82.7	0.87
Stimulus intensity: tDCS-Ca vs Sham	−8.86e-03	0.8	1	69.9	0.37
Trials: tDCS-Ca vs Sham	−3.48e-03	1.4	1	94.2	0.23
Days:runs:trials	−9.40e-04	0.3	1	93.2	0.59
Runs:trials: tDCS-An vs other	5.66e-05	<0.001	1	125.2	0.98
Runs:trials:tDCS-Ca vs Sham	1.43e-03	0.2	1	134.7	0.65

* $P < 0.05$ two-tailed.† $P < 0.001$ two-tailed.

‡ n.a., not applicable.

For Stim Order, a categorical variable with 6 levels, only the overall F-test and P value are reported; no single estimate is shown.

outcomes and adjust their responses accordingly (ie, participants reporting what they think experimenters expect them to report),⁷¹ a factor of particular relevance in pain research where expectations strongly influence reported experience.⁴⁷ Furthermore, we conducted systematic assessments of blinding, sensitivity to modeling choices and inclusion of covariates, and reaction time. These enhancements helped produce a larger and more reliable effect size than many earlier studies, addressing key gaps in reproducibility and methodological consistency in the tDCS pain literature.

The results demonstrated that anodal tDCS applied to left M1 significantly reduced both low and high pain intensity ratings for trials administered to the contralateral right forearm, while cathodal tDCS increased low pain intensity, when comparing pre- and post-tDCS pain ratings. In addition, reaction time by the left hand in response to pain stimuli to the right forearm was significantly increased after anodal stimulation, suggesting possible interhemispheric inhibition effects. No significant interaction was observed between tDCS and runs or trials, indicating consistent effects across time. The analgesic effect of anodal tDCS and the hyperalgesic effect of cathodal tDCS remained robust throughout the poststimulation testing period. Notably, across all conditions, pain ratings dropped sharply from the first to the second trial (**Fig. 4**). The sharp drop in pain ratings

from the first to the second trial across all 3 runs and in both pre- and post-tDCS sessions may reflect decreased nociceptor responsiveness and reduced central pain processing, as well as heightened initial responses due to novelty or anticipatory arousal that fade with repeated exposure.^{46,47}

The analgesic effects of anodal M1-tDCS may reflect engagement of descending pain modulatory pathways and increased inhibitory control over pain-processing regions. Pre-clinical studies have implicated brainstem structures such as the PAG and RVM in pain modulation after M1 stimulation,³⁴ along with modulation of thalamocortical circuits,^{52,69} which could further affect pain perception. Conversely, cathodal tDCS may reduce cortical excitability and disrupt these pathways, leading to increased pain perception.⁵⁵ These circuit-level mechanisms remain speculative, as they were not directly assessed in this study.

The observed slowing of reaction time in the left hand after anodal stimulation of the left M1 is consistent with interhemispheric inhibition, where increased excitability in the stimulated hemisphere suppresses contralateral motor cortex activity. This phenomenon has been demonstrated in previous motor cortex stimulation studies.^{29,84,85} The corpus callosum likely mediates this inhibitory interaction, facilitating suppression of contralateral activity to optimize task performance and support hemispheric

specialization.^{8,16,51} While interhemispheric inhibition was not directly measured in this study, it can be assessed using TMS-based protocols or concurrent neuroimaging techniques in future studies.

The results of this study have potential clinical implications, particularly for the use of tDCS in the treatment of different pain conditions. While experimental pain and clinical pain are not the same phenomena, evoked hypersensitivity and/or allodynia is a feature of many chronic pain conditions^{86,87} and an important part of the broader syndrome of chronic pain, energetic, and mood dysregulation experienced by many people in chronic pain. Accordingly, quantitative sensory testing of evoked pain responses is an important, widely used tool for helping to diagnose and understand chronic pain.^{3,59,77} In addition, protocols similar to the M1 stimulation that we studied here have been used to modulate clinical pain, both with tDCS in research studies^{4,58,62,81} and TMS in research studies and clinical practice.^{14,35,81} The ability to modulate pain perception non-invasively through targeted cortical stimulation holds significant promise for improving pain management, especially in patients who do not respond well to traditional pharmacological treatments. The findings from this study suggest that tDCS could be effective in reducing pain at a fundamental level, even when it is known to be nociceptive in origin. Furthermore, this study demonstrates evidence for experimental pain modulation, but more work is needed to evaluate its efficacy in clinical populations. Future studies should test this approach in individuals with chronic pain to determine its translational potential. Although the present findings demonstrate that M1-tDCS can modulate pain perception behaviorally, the underlying neural mechanisms remain speculative and were not directly assessed in this study. Future research should use neuroimaging techniques such as functional MRI (fMRI) to assess activity changes in the PAG, thalamus, and pain-related cortical regions, as well as electroencephalography (EEG) to evaluate cortical excitability and interhemispheric connectivity. These approaches would help determine whether M1-tDCS engages descending modulatory systems, influences thalamo-cortical loops, or modulates sensorimotor integration. To address durability, we explicitly tested the interaction of tDCS Type with both Runs and Trials in our linear mixed-effects model (equation 1). As summarized in **Table 1**, these interactions were nonsignificant, suggesting that the effects of tDCS remained stable across time without evidence of decay across the 35-minute poststimulation period. Further research is also needed to refine stimulation protocols, test durability (eg, 24 hours vs 1 month), optimize parameters, and test generalizability across different body sites, affective pain components, and chronic pain conditions. Establishing the neural mechanisms of tDCS in humans is critical for translating these findings into effective and personalized pain treatments.

One limitation of this study is the use of standard electrode placement to target the left M1 without accounting for individual anatomical variability. Although C3 is commonly used as a proxy for M1 in tDCS studies, this method does not guarantee precise localization across participants due to interindividual differences in cortical anatomy.^{18,43,54,66,70} As such, it is possible that stimulation extended beyond or did not optimally target M1 in all individuals. This limitation should be considered when interpreting the observed effects as M1-specific. Future studies using MRI-guided individualized current flow modeling would provide more accurate targeting and help clarify the specific cortical regions involved in mediating the effects of tDCS. Another limitation is the exclusion of participants whose pain threshold and tolerance differed by less than 1°C. This criterion, consistent

with prior experimental pain research,^{27,72} was applied for both scientific and safety reasons. A minimum 1°C gap ensured that the “low” and “high” temperatures were perceptually distinct, enabling valid comparisons of pain modulation while minimizing the risk of overstimulation or tissue damage. Importantly, previous reliability studies of quantitative sensory testing have shown that the standard error of measurement for heat pain thresholds is approximately 1°C, meaning that smaller differences fall within typical measurement error and cannot be distinguished.⁶⁴ Classic psychophysical validation studies have further demonstrated that pain ratings behave as ratio-scale measures across clearly separated stimulus intensities, reinforcing the need for perceptually distinct stimulus differences.⁷⁶ Future research aimed at clinical translation should evaluate neurostimulation effects in larger, more diverse populations that reflect the communities they are intended to serve.

5. Conclusions

This study demonstrates that 20 minutes of anodal tDCS over the left primary motor cortex produces large, durable reductions in evoked nociceptive pain ($P < 0.001$; Cohen's $d = 0.93$ and 0.85 compared with cathodal and sham, respectively), and cathodal stimulation increased pain relative to sham ($P < 0.01$; Cohen's $d = 0.34$). The polarity-specific effects were robust and remained stable across a 35-minute poststimulation testing period. Conducted as a pre-registered, triple-blind, randomized crossover trial with a larger sample size ($N = 50$), the study overcame key limitations of prior work through methodological rigor and comprehensive experimental control. We used a within-subject design to reduce interindividual variability, an adaptive pain calibration procedure to personalize stimulus intensity, and dynamic pain assessment across multiple trials to enhance sensitivity to change. The inclusion of anodal, cathodal, and sham conditions in counterbalanced order, with sufficient washout periods, minimized carryover effects and enabled strong within-participant contrasts. Importantly, the effects were unrelated to tDCS-evoked sensations, and systematic blinding assessments confirmed that participants were unable to reliably distinguish active from sham stimulation, strengthening internal validity. Reaction time data further revealed a modest slowing after anodal stimulation ($P < 0.05$; Cohen's $d = 0.16$), consistent with interhemispheric motor inhibition, although this change was uncorrelated with analgesia, suggesting independent effects on sensory and motor dimensions of pain processing. Taken together, these findings provide compelling, reproducible evidence that M1-tDCS produces polarity-specific modulation of pain and highlight the value of rigorous design for advancing noninvasive neuro-modulation research.

Conflict of interest statement

The authors have no conflicts of interest to declare. MAN is on the Scientific Advisory Boards of Neuroelectronics, Inc. and Precisis, Inc.

Acknowledgments

Supported by the National Institute of Mental Health (No. R37 MH076136) and the Neural Code Cluster at Dartmouth College. All data used for analyses in this study are publicly available in the CANLAB repository at https://github.com/canlab/stimMAP_2025#.

Article history:

Received 7 May 2025

Received in revised form 23 September 2025

Accepted 4 October 2025

Available online 4 December 2025

References

- [1] Angius L, Hopker JG, Marcora SM, Mauger AR. The effect of transcranial direct current stimulation of the motor cortex on exercise-induced pain. *Eur J Appl Physiol* 2015;115:2311–9.
- [2] Antal A, Brepohl N, Poreisz C, Boros K, Csifcsak G, Paulus W. Transcranial direct current stimulation over somatosensory cortex decreases experimentally induced acute pain perception. *Clin J Pain* 2008;24:56–63.
- [3] Backonja M-M, Walk D, Edwards RR, Sehgal N, Moeller-Bertram T, Wasan A, Irving G, Argoff C, Wallace M. Quantitative sensory testing in measurement of neuropathic pain phenomena and other sensory abnormalities. *Clin J Pain* 2009;25:641–7.
- [4] Bai Y, Pacheco-Barrios K, Pacheco-Barrios N, Liang G, Fregni F. Neurocircuitry basis of motor cortex-related analgesia as an emerging approach for chronic pain management. *Nat Ment Health* 2024;2:496–513.
- [5] Bartoshuk LM, Duffy VB, Green BG, Hoffman HJ, Ko C-W, Lucchina LA, Marks LE, Snyder DJ, Weiffenbach JM. Valid across-group comparisons with labeled scales: the gLMS versus magnitude matching. *Physiol Behav* 2004;82:109–14.
- [6] Biabani M, Farrell M, Zoghi M, Egan G, Jaberzadeh S. Crossover design in transcranial direct current stimulation studies on motor learning: potential pitfalls and difficulties in interpretation of findings. *Rev Neurosciences* 2018;29:463–73.
- [7] Bikson M, Grossman P, Thomas C, Zannou AL, Jiang J, Adnan T, Mourdoukoutas AP, Kronberg G, Truong D, Boggio P, Brunoni AR, Charvet L, Fregni F, Fritsch B, Gillick B, Hamilton RH, Hampstead BM, Jankord R, Kirtan A, Knotkova H, Liebetanz D, Liu A, Loo C, Nitsche MA, Reis J, Richardson JD, Rotenberg A, Turkeltaub PE, Woods AJ. Safety of transcranial direct current stimulation: evidence based update 2016. *Brain Stimulation* 2016;9:641–61.
- [8] Bloom JS, Hynd GW. The role of the corpus callosum in interhemispheric transfer of information: excitation or inhibition? *Neuropsychol Rev* 2005;15:59–71.
- [9] Boggio PS, Zaghi S, Fregni F. Modulation of emotions associated with images of human pain using anodal transcranial direct current stimulation (tDCS). *Neuropsychologia* 2009;47:212–7.
- [10] Borckardt JJ, Bikson M, Frohman H, Reeves ST, Datta A, Bansal V, Madan A, Barth K, George MS. A pilot study of the tolerability and effects of high-definition transcranial direct current stimulation (HD-tDCS) on pain perception. *J Pain* 2012;13:112–20.
- [11] Brasil-Neto JP, Iannone A, Caixeta FV, Cavendish BA, de Mello Cruz AP, Buratto LG. Acute offline transcranial direct current stimulation does not change pain or anxiety produced by the cold pressor test. *Neurosci Lett* 2020;736:135300.
- [12] Cai G, Xia Z, Charvet L, Xiao F, Datta A, Androulakis XM. A systematic review and meta-analysis on the efficacy of repeated transcranial direct current stimulation for migraine. *J Pain Res* 2021;14:1171–83.
- [13] Caumo W, Franca BR, Orzechowski R, Bueno G, França de Souza A, Dos Santos da Silva JV, Sanches PRS, Da Silva DP Jr, Torres ILS, Hirakata VN, Pacheco-Barrios K, Fregni F. Home-based transcranial direct current stimulation vs placebo for fibromyalgia: a randomized clinical trial. *JAMA Netw Open* 2025;8:e2514262.
- [14] Cavaleri R, Chipchase LS, Summers SJ, Schabrun SM. Repetitive transcranial magnetic stimulation of the primary motor cortex expedites recovery in the transition from acute to sustained experimental pain: a randomised, controlled study. *PAIN* 2019;160:2624–33.
- [15] Cheng Y-C, Hsiao C-Y, Su M-I, Chiu C-C, Huang Y-C, Huang W-L. Treating fibromyalgia with electrical neuromodulation: a systematic review and meta-analysis. *Clin Neurophysiol* 2023;148:17–28.
- [16] Cook ND. Homotopic callosal inhibition. *Brain Lang* 1984;23:116–25.
- [17] Dasilva AF, Mendonca ME, Zaghi S, Lopes M, Dossantos MF, Spierings EL, Bajwa Z, Datta A, Bikson M, Fregni F. tDCS-induced analgesia and electrical fields in pain-related neural networks in chronic migraine. *Headache* 2012;52:1283–95.
- [18] Datta A, Truong D, Minhas P, Parra LC, Bikson M. Inter-individual variation during transcranial direct current stimulation and normalization of dose using MRI-derived computational models. *Front Psychiatry* 2012;3:91.
- [19] David MCM, Moraes AA, Costa ML, Franco CIF. Transcranial direct current stimulation in the modulation of neuropathic pain: a systematic review. *Neurol Res* 2018;40:557–65.
- [20] Dayan E, Censor N, Buch ER, Sandrini M, Cohen LG. Noninvasive brain stimulation: from physiology to network dynamics and back. *Nat Neurosci* 2013;16:838–44.
- [21] Dehghani A, Bango C, Murphy EK, Halter RJ, Wager TD. Independent effects of transcranial direct current stimulation and social influence on pain. *PAIN* 2025;166:87–98.
- [22] Deldar Z, Rustamov N, Blanchette I, Piché M. Improving working memory and pain inhibition in older persons using transcranial direct current stimulation. *Neurosci Res* 2019;148:19–27.
- [23] De Ridder D, Vanneste S. Occipital nerve field transcranial direct current stimulation normalizes imbalance between pain detecting and pain inhibitory pathways in fibromyalgia. *Neurotherapeutics* 2017;14:484–501.
- [24] DosSantos MF, Love TM, Martikainen IK, Nascimento TD, Fregni F, Cummiford C, Deboer MD, Zubieta J-K, Dasilva AFM. Immediate effects of tDCS on the μ -opioid system of a chronic pain patient. *Front Psychiatry* 2012;3:93.
- [25] DosSantos MF, Martikainen IK, Nascimento TD, Love TM, DeBoer MD, Schambra HM, Bikson M, Zubieta J-K, DaSilva AF. Building up analgesia in humans via the endogenous μ -opioid system by combining placebo and active tDCS: a preliminary report. *PLoS One* 2014;9:e102350.
- [26] DosSantos MF, Oliveira AT, Ferreira NR, Carvalho ACP, Rosado de Castro PH. The contribution of endogenous modulatory systems to TMS- and tDCS-induced analgesia: evidence from PET studies. *Pain Res Manage* 2018;2018:1–14.
- [27] Edwards RR, Fillingim RB. Self-reported pain sensitivity: lack of correlation with pain threshold and tolerance. *Eur J Pain* 2007;11:594–8.
- [28] Fenton BW, Palmieri PA, Boggio P, Fanning J, Fregni F. A preliminary study of transcranial direct current stimulation for the treatment of refractory chronic pelvic pain. *Brain Stimulation* 2009;2:103–7.
- [29] Ferbert A, Priori A, Rothwell JC, Day BL, Colebatch JG, Marsden CD. Interhemispheric inhibition of the human motor cortex. *J Physiol* 1992;453:525–46.
- [30] Fillingim RB, King CD, Ribeiro-Dasilva MC, Rahim-Williams B, Riley JL III. Sex, gender, and pain: a review of recent clinical and experimental findings. *J Pain* 2009;10:447–85.
- [31] Flood A, Waddington G, Keegan RJ, Thompson KG, Cathcart S. The effects of elevated pain inhibition on endurance exercise performance. *PeerJ* 2017;5:e3028.
- [32] Fregni F, Gimenes R, Valle AC, Ferreira MJL, Rocha RR, Natalle L, Bravo R, Rigonatti SP, Freedman SD, Nitsche MA, Pascual-Leone A, Boggio PS. A randomized, sham-controlled, proof of principle study of transcranial direct current stimulation for the treatment of pain in fibromyalgia. *Arthritis Rheum* 2006;54:3988–98.
- [33] Gantz D, Dehghani A, Wager T. Investigating the impact of transcranial direct current stimulation (tDCS) on pain perception, altruism, and placebo effects. 2024.
- [34] Gan Z, Li H, Naser PV, Oswald MJ, Kuner R. Suppression of neuropathic pain and comorbidities by recurrent cycles of repetitive transcranial direct current motor cortex stimulation in mice. *Sci Rep* 2021;11:9735.
- [35] Gatzinsky K, Bergh C, Liljgren A, Silander H, Samuelsson J, Svanberg T, Samuelsson O. Repetitive transcranial magnetic stimulation of the primary motor cortex in management of chronic neuropathic pain: a systematic review. *Scand J Pain* 2021;21:8–21.
- [36] Glaser J, Reeves ST, Stoll WD, Epperson TI, Hilbert M, Madan A, George MS, Borckardt JJ. Motor/prefrontal transcranial direct current stimulation (tDCS) following lumbar surgery reduces postoperative analgesia use. *Spine (Phila PA 1976)* 2016;41:835–9.
- [37] Gregoret L, Zamorano AM, Graven-Nielsen T. Effects of multifocal transcranial direct current stimulation targeting the motor network during prolonged experimental pain. *Eur J Pain* 2021;25:1241–53.
- [38] Grundmann L, Rolke R, Nitsche MA, Pavlakovic G, Happe S, Treede R-D, Paulus W, Bachmann CG. Effects of transcranial direct current stimulation of the primary sensory cortex on somatosensory perception. *Brain Stimulation* 2011;4:253–60.
- [39] Gurdíel-Álvarez F, González-Zamorano Y, Lerma Lara S, Gómez-Soriano J, Taylor J, Romero JP, Gómez Jiménez M, Fernández-Carnero J. Effectiveness of unihemispheric concurrent dual-site stimulation over M1 and dorsolateral prefrontal cortex stimulation on pain processing: a triple blind cross-over control trial. *Brain Sci* 2021;11:188.
- [40] Haikalis NK, Hooyman A, Wang P, Daliri A, Schaefer SY. Placebo effects of transcranial direct current stimulation on motor skill acquisition. *Neurosci Lett* 2023;814:137442.
- [41] Hammer JW, Villamar MF, Fregni F, Taylor JA. Transcranial direct current stimulation (tDCS) and the cardiovascular responses to acute pain in humans. *Clin Neurophysiol* 2015;126:1039–46.
- [42] Hashmi JA, Davis KD. Effects of temperature on heat pain adaptation and habituation in men and women. *PAIN* 2010;151:737–43.

- [43] Herwig U, Satrapi P, Schönfeldt-Lecuona C. Using the international 10-20 EEG system for positioning of transcranial magnetic stimulation. *Brain Topogr* 2003;16:95–9.
- [44] Hollins M, Harper D, Maixner W. Changes in pain from a repetitive thermal stimulus: the roles of adaptation and sensitization. *PAIN* 2011;152:1583–90.
- [45] Hughes SW, Ali M, Sharma P, Insan N, Strutton PH. Frequency-dependent top-down modulation of temporal summation by anodal transcranial direct-current stimulation of the primary motor cortex in healthy adults. *Eur J Pain* 2018;22:1494–501.
- [46] Jepma M, Jones M, Wager TD. The dynamics of pain: evidence for simultaneous site-specific habituation and site-nonspecific sensitization in thermal pain. *J Pain* 2014;15:734–46.
- [47] Jepma M, Wager TD. Conceptual conditioning: mechanisms mediating conditioning effects on pain. *Psychol Sci* 2015;26:1728–39.
- [48] Jepma M, Wager TD. Multiple potential mechanisms for context effects on pain. *PAIN* 2013;154:629–31.
- [49] Jiang X, Wang Y, Wan R, Feng B, Zhang Z, Lin Y, Wang Y. The effect of high-definition transcranial direct current stimulation on pain processing in a healthy population: a single-blinded crossover controlled study. *Neurosci Lett* 2022;767:136304.
- [50] Jürgens TP, Schulte A, Klein T, May A. Transcranial direct current stimulation does neither modulate results of a quantitative sensory testing protocol nor ratings of suprathreshold heat stimuli in healthy volunteers. *Eur J Pain* 2012;16:1251–63.
- [51] Kinsbourne M. Hemispheric specialization and the growth of human understanding. *Am Psychol* 1982;37:411–20.
- [52] Knotkova H, Nitsche MA, Cruciani RA. Putative physiological mechanisms underlying tDCS analgesic effects. *Front Hum Neurosci* 2013;7:628.
- [53] Kold S, Kragh AJ, Graven-Nielsen CS, Elnegaard FS, Lund F, Vittrup IV, Cliff KL, Sivarooban R, Petrini L. Neuromodulation of somatosensory pain thresholds of the neck musculature using a novel transcranial direct current stimulation montage: a randomized double-blind, sham controlled study. *Scand J Pain* 2022;22:622–30.
- [54] Laakso I, Tanaka S, Mikkonen M, Koyama S, Sadato N, Hirata A. Electric fields of motor and frontal tDCS in a standard brain space: a computer simulation study. *Neuroimage* 2016;137:140–51.
- [55] Lefaucheur J-P, Wendling F. Mechanisms of action of tDCS: a brief and practical overview. *Neurophysiol Clin* 2019;49:269–75.
- [56] Lim M, Kim DJ, Nascimento TD, DaSilva AF. High-definition tDCS over primary motor cortex modulates brain signal variability and functional connectivity in episodic migraine. *Clin Neurophysiol* 2024;161:101–11.
- [57] Lin RL, Douaud G, Filippini N, Okell TW, Stagg CJ, Tracey I. Structural connectivity variances underlie functional and behavioral changes during pain relief induced by neuromodulation. *Sci Rep* 2017;7:41603.
- [58] Li X, Yao J, Zhang W, Chen S, Peng W. Effects of transcranial direct current stimulation on experimental pain perception: a systematic review and meta-analysis. *Clin Neurophysiol* 2021;132:2163–75.
- [59] Maixner W, Fillingim RB, Williams DA, Smith SB, Slade GD. Overlapping chronic pain conditions: implications for diagnosis and classification. *J Pain* 2016;17:T93–T107.
- [60] Mariano TY, Van't Wout M, Garnaat SL, Rasmussen SA, Greenberg BD. Transcranial direct current stimulation (tDCS) targeting left dorsolateral prefrontal cortex modulates task-induced acute pain in healthy volunteers. *Pain Med* 2016;17:737–45.
- [61] Mechsner S, Grünert J, Wiese JJ, Vormbäumen J, Sehouli J, Siegmund B, Neeb L, Prüß MS. Transcranial direct current stimulation to reduce chronic pelvic pain in endometriosis: phase II randomized controlled clinical trial. *Pain Med* 2023;24:809–17.
- [62] Meeker TJ, Keaser ML, Khan SA, Gullapalli RP, Seminowicz DA, Greenspan JD. Non-invasive motor cortex neuromodulation reduces secondary hyperalgesia and enhances activation of the descending pain modulatory network. *Front Neurosci* 2019;13:467.
- [63] Mehta S, McIntyre A, Guy S, Teasell RW, Loh E. Effectiveness of transcranial direct current stimulation for the management of neuropathic pain after spinal cord injury: a meta-analysis. *Spinal Cord* 2015;53:780–5.
- [64] Middlebrook N, Heneghan NR, Evans DW, Rushton A, Falla D. Reliability of temporal summation, thermal and pressure pain thresholds in a healthy cohort and musculoskeletal trauma population. *PLoS One* 2020;15:e0233521.
- [65] van der Miesen MM, Joosten EA, Kaas AL, Linden DEJ, Peters JC, Vossen CJ. Habituation to pain: self-report, electroencephalography, and functional magnetic resonance imaging in healthy individuals. A scoping review and future recommendations. *PAIN* 2024;165:500–22.
- [66] Miranda PC, Lomarev M, Hallett M. Modeling the current distribution during transcranial direct current stimulation. *Clin Neurophysiol* 2006;117:1623–9.
- [67] Mordillo-Mateos L, Dileone M, Soto-León V, Brocalero-Camacho A, Pérez-Borrego YA, Onate-Figueroa A, Aguilar J, Oliviero A. Effects of transcranial direct current stimulation on temperature and pain perception. *Sci Rep* 2017;7:2946.
- [68] Nakagawa K, Koyama S, Inui K, Tanaka S, Kakigi R, Sadato N. Polarity-independent effects of transcranial direct current stimulation over the bilateral opercular somatosensory region: a magnetoencephalography study. *Neuroreport* 2017;28:838–44.
- [69] Nitsche MA, Paulus W. Excitability changes induced in the human motor cortex by weak transcranial direct current stimulation. *J Physiol* 2000;527:633–9.
- [70] Opitz A, Paulus W, Will S, Antunes A, Thielscher A. Determinants of the electric field during transcranial direct current stimulation. *Neuroimage* 2015;109:140–50.
- [71] Orne MT. On the social psychology of the psychological experiment: with particular reference to demand characteristics and their implications. *American Psychologist* 1962;17:776–783.
- [72] Parsons J, Delity MJ, Brookes J, Akintola T, Dildine T, Owens H, Barhoum S, Yanes JA, Atlas LY. Heat pain threshold and tolerance do not depend on thermode type or stimulus duration. *Psychology* 2024;72:10.
- [73] Pereira M, Rafiq B, Chowdhury E, Babayev J, Boo H, Metwally R, Sandilya P, Chusid E, Battaglia F. Anodal cerebellar tDCS modulates lower extremity pain perception. *NeuroRehabilitation* 2017;40:195–200.
- [74] Pinto CB, Teixeira Costa B, Duarte D, Fregni F. Transcranial direct current stimulation as a therapeutic tool for chronic pain. *J ECT* 2018;34:e36–e50.
- [75] Powers A, Madan A, Hilbert M, Reeves ST, George M, Nash MR, Borckardt JJ. Effects of combining a brief cognitive intervention with transcranial direct current stimulation on pain tolerance: a randomized controlled pilot study. *Pain Med* 2018;19:677–85.
- [76] Price D, McGrath P, Rafii A, Buckingham B. The validation of visual analogue scales as ratio scale measures for chronic and experimental pain. *PAIN* 1983;17:45–56.
- [77] Rolke R, Baron R, Maier C, Tölle TR, Treede DR, Beyer A, Binder A, Birbaumer N, Birklein F, Bötefür IC, Braune S, Flor H, Hoge V, Klug R, Landwehrmeyer GB, Magerl W, Maihöfner C, Rolko C, Schaub C, Scherens A, Sprenger T, Valet M, Wasserka B. Quantitative sensory testing in the German Research Network on Neuropathic Pain (DFNS): standardized protocol and reference values. *PAIN* 2006;123:231–43.
- [78] Schambra HM, Bikson M, Wager TD, DosSantos MF, DaSilva AF. It's all in your head: reinforcing the placebo response with tDCS. *Brain Stimul* 2014;7:623–4.
- [79] Schenk LA, Fadaei T, Büchel C. How side effects can improve treatment efficacy: a randomized trial. *Brain* 2024;147:2643–51.
- [80] Serrano PV, Zortea M, Alves RL, Beltrán G, Bavaresco C, Ramalho L, Alves CFdS, Medeiros L, Sanches PRS, Silva DP Jr, Lucena da Silva Torres I, Fregni F, Caumo W. The effect of home-based transcranial direct current stimulation in cognitive performance in fibromyalgia: a randomized, double-blind sham-controlled trial. *Front Hum Neurosci* 2022;16:992742.
- [81] Silva VA, Baptista AF, Fonseca AS, Carneiro AM, Brunoni AR, Carrilho PEM, Lins CC, Kubota GT, Fernandes AMBL, Lapa JDS, Dos Santos LM, Sasso I, Monte-Silva K, Poindessous-Jazat F, Mori N, Miki K, Baltar A, Tanaka C, Teixeira MJ, Hosomi K, Bouhassira D, Attal N, Ciampi de Andrade D. Motor cortex repetitive transcranial magnetic stimulation in fibromyalgia: a multicentre randomised controlled trial. *Br J Anaesth* 2025;134:1756–1764.
- [82] Stacheneder R, Alt L, Straube A, Ruscheweyh R. Effects of transcranial direct current stimulation (t-DCS) of the cerebellum on pain perception and endogenous pain modulation: a randomized, monocentric, double-blind, sham-controlled crossover study. *Cerebellum* 2022;22:1234–42.
- [83] Stagg CJ, Nitsche MA. Physiological basis of transcranial direct current stimulation. *Neuroscientist* 2011;17:37–53.
- [84] Tazoe T, Endoh T, Kitamura T, Ogata T. Polarity specific effects of transcranial direct current stimulation on interhemispheric inhibition. *PLoS One* 2014;9:e114244.
- [85] Wilkins EW, Young RJ, Houston D, Kawana E, Lopez Mora E, Sunkara MS, Riley ZA, Poston B. Non-dominant hemisphere excitability is unaffected during and after transcranial direct current stimulation of the dominant hemisphere. *Brain Sci* 2024;14:694.
- [86] Woolf CJ. Evidence for a central component of post-injury pain hypersensitivity. *Nature* 1983;306:686–8.
- [87] Woolf CJ, Safieh-Garabedian B, Ma QP, Crilly P, Winter J. Nerve growth factor contributes to the generation of inflammatory sensory hypersensitivity. *Neuroscience* 1994;62:327–31.
- [88] Xiong H-Y, Cao Y-Q, Du S-H, Yang Q-H, He S-Y, Wang X-Q. Effects of high-definition transcranial direct current stimulation targeting the anterior cingulate cortex on the pain thresholds: a randomized controlled trial. *Pain Med* 2023;24:89–98.